



Water Information
DATA › INFORMATION › INSIGHT

BoM Hydro Field Event Message Format

R.Thompson

August 2010

Version 1.0



Australian Government
Bureau of Meteorology

www.bom.gov.au/water

Revision history

Date	Version	Description	Author
16/8/2010	1.0	Formal release	R.Thompson

Release history

Date	Version	Status	Audience	Approval
16/8/2010	1.0	First Release includes "HS=1" Event	BoM	R.Thompson

Document management register

Document	File reference

Review Status

Reviewer	Date Reviewed	Version Reviewed
L.Incani		0.1,0.2,0.3
G.Shaw		0.2,0.3
D.Chong		0.3
J.Robinson		0.3
W.Shallcross		0.3

Release Signatories

Approval	Name	Signature	Date

Contact details

Robert Thompson

Bureau of Meteorology
GPO Box 1289 MELBOURNE VIC 3001

Phone:
Email: R.Thompson@bom.gov.au

Contents

1	Introduction	6
2	Message Syntax	7
2.1	HFEM Structure	7
2.2	Message Header Section	7
2.3	Maintenance Mode Section	8
2.4	Message Body	8
2.5	Message Footer Section	8
3	Message Types	9
4	Message Body – Hydro System (HS), version 1	10
4.1	Site Identifier Parameter	10
4.2	Observation Timestamp Parameter	11
4.3	Sensor Measurement Parameters	13
4.4	Coded “HS=1” Message Examples	17
	References	19

List of Tables

Table 1 - HFEM Defined Message Types..... 9

Table 2 - Site Identification Schemes10

Table 3 - Observation Time Schemes.....11

Table 4 - Sensor Type Parameters14

Table 5 - Sensor Measurement Types15

1 Introduction

The Bureau of Meteorology Hydrology field event message (HFEM) has been designed for use with field data acquisition devices (data loggers) that have communications attached and are capable of sending (initiating) messages based on a defined sensor equipment trigger or inactivity period.

HFEM uses a *name-value* pair encoding method with ASCII character representation. This encoding method provides for compact messages to fit within required carrier payloads. The message format is designed for and intended for an alert based transmission methodology for event (triggered) sensors, this type of data on-forwarding is best suited for real-time environments such as flood and hydrological monitoring.

The HFEM format also has the advantage that it is easily human readable allowing for diagnostics at key stages in the transmission stream without the need for special binary message decoders and viewing tools. Further message size reduction could be accomplished by binary encoding and would be recommended for direct radio communications channels, however with this approach it draws in other complexities such as Endianness and readability. HFEM is well suited to communications over modern IP and session based communications. XML formatted messages are excluded from this encoding scheme as they are considered not suitable for the intended use in which HFEM will be deployed. Light weight transfer protocols are required for field telemetry, XML is considered too verbose it will add far greater communication and processing overheads through increased message charges and increased processing on embedded controllers. Maximum payload sizes also need to be carefully managed as XML adds significant overhead due to its structure.

The HFEM format is a fixed outer wrapper and will never change, the body of the message will change dependant on the message type and version that is used.

2 Message Syntax

The HFEM structure consists of a start of message (SOM), the message itself and terminated by an end of message (EOM). The SOM and EOM are identified by the colon (':') character. Sections and named-value pairs within the message are separated by the pipe '|' character.

Message Syntax,

:{MESSAGE}:

Syntax Notes:

1. This document identifies mandatory message sections within curly brackets "{ }", optional sections are enclosed within square brackets "[]".
2. For message decoding purposes numeric fields do not require leading zeros unless the parameter or value mandates it. If this is the case it should be clearly specified for that message type. E.g. '000123' would be treated the same as '123' unless the parameter format mandates fixed formatting.
3. Spaces are considered white space and do not have any significance within the message. White space should be avoided to keep messages compact.

2.1 HFEM Structure

The Bureau HFEM is made up of a header section, optional maintenance mode section, section body of which the contents is dependant on the message type and a footer section. As outlined above these message parts are wrapped by an SOM and EOM.

The HFEM is based on name-value pairs, as such the order of the parameters within the body can be in any order as each named parameter is unique. It is mandatory to always start with a SOM then a header and optionally followed by the maintenance section. The message body section will vary in contents and length as these characteristics are determined by the message type and version (described later in the document). The message body is always closed by a footer section and terminated by an EOM.

Syntax for the message section is,

MESSAGE: {HEADER} | [MAINTMODE] | {MESSAGE-TYPE-BODY} | {FOOTER}

2.2 Message Header Section

The header section identifies the 'message type' to the receiving system, typically a message decoding system or message switch. To assist in the maintenance and

extensibility of the message format a message sequence (version) number is added to the header as the value part inline with the name-value pair encoding. The sequence (version) number will be incremented with each update (change) in the 'message type' structure. The message type sent will be accordance to the table in section 3.

Syntax for this section is,

HEADER: {XX=nn}

2.3 Maintenance Mode Section

The maintenance mode section is an optional message section and used by the logging device to indicate the message is being sent while the logger is in maintenance mode. This mode suggests the enclosed data should not be used in production but rather as a check message for field testing and communications path validation. This mode would typically be evoked by the field technician to send a field test message.

Syntax for this section is,

MAINTMODE: [M=1]

2.4 Message Body

The message body will vary in content as it is dependant on the message type (section 3), however components within the body will always be separated by the pipe '|' character.

Even though theoretically the parameter component parts within a (message type) body can be arranged in any order due to the uniqueness of named parameters, the system implementation may enforce a specific order so caution should be applied. It is suggested to follow the order that is outlined within each message type section and the specific version you wish to implement.

2.5 Message Footer Section

The footer section identifies the end of parameters for this message been reached. It is the only parameter that does not have a paired value as it is a special case. The footer 'NN' and EOM uniquely identify that the message has ended.

Syntax for this section is,

FOOTER: {NN}

3 Message Types

The message type is used to describe in the message header what sort of contents this message sends, these message types will be updated as need demands.

Table 1 - HFEM Defined Message Types

Message Type	Message Type Name	Description
BoM Hydro System Event	HS	Hydrological system field equipment that generate events and initiate the message push to the destination system/s.

4 Message Body – Hydro System (HS), version 1

The following section describes the parameters within the message body to be used by a 'hydro system' event message type.

As per table 1 the hydro system event type is denoted by the "HS" message type code. The 'HS' body section starts with a site identifier parameter, optional observation time stamp which is associated to the sensor measurement or measurements that follow. Each message may report one or sensor measurements with each name-valued pair being separated by the pipe '|' character as per the message syntax standard.

Syntax for this section is,

MESSAGE-TYPE-BODY:

{SITEID}[|OBSTIME|]{SENSORMEASURE}[|...|{SENSORMEASURE}]

The following document sections describe the individual parameters within the Hydro System event message body.

4.1 Site Identifier Parameter

The SITE-ID parameter identifies the site where this message originates from. Different site identification schemes are possible and may be extended in later versions, if required. The site identification tag will take the form 'I' with an appended scheme ID number.

Syntax for this parameter is,

SITEID: {In='value format as per table below'}

Table 2 - Site Identification Schemes

Site ID Scheme Type	Site Parameter Name	Value Format	Description
Site Identifier	I1	XXXXXXXX	Official site identifier as provided from sites database. Up to eight alphanumeric identifier. e.g. "12345678"
External Agency Site Identifier	I2	XXXXXX-XXXXXXX	Used when sending messages to an alternate agency and an external agency identifier is required. The external identifier is a six character alphanumeric and a hyphenated

			eight character alphanumeric site identifier. e.g. BOM002-12345678
--	--	--	---

4.2 Observation Timestamp Parameter

The Timestamp section is an optional but highly recommended section. The observation timestamp section is the timestamp for each of the accompanying sensor measurement parameters within the message body. If a sensor measurement has different observation times they must be sent as separate messages to ensure time and value accuracy.

Three timestamp schemes are supported, if required additional formats will be added in later versions. The Bureau standard is to send observations with a UTC time format, this will eliminate the need for site time zone metadata at each of the data ingestion systems as all data observations are stored in UTC format.

Syntax for this parameter is,

OBSTIME: [Tn='value format as per table below']

Table 3 - Observation Time Schemes

Timestamp Scheme Type	Timestamp Parameter Name	Value Format	Description
UTC Type 1	T1	CCYYMMDDHHMISS	Observation timestamp sent as universal time coordinate (UTC). The hours component is in 24 hour mode. e.g. 1 st July 2010 at 1pm and 10 seconds will be: 20100701130010
UTC Type 2	T2	CCYYMMDDHHMISS{+/-}TZ	Observation timestamp sent as universal time coordinate (UTC). The hours

			component is in 24 hour mode and the time zone offset is appended with appropriate sign to yield a <u>calculated</u> local standard time. e.g. UTC time of 1st July 2010 at 1pm and 10 seconds, sent from a site in Victoria/Australia which has a standard time offset from UTC of +10 hours would be represented as: 20100701130010+10
LST - Local Standard Time.	T3	CCYYMMDDHHMISS{+/-}TZ	Observation timestamp sent as local "standard" time. The hours component is in 24 hour mode and the time zone offset is appended with appropriate sign to yield a <u>calculated</u> UTC time. e.g. LST time of 1st July 2010 at 3 am and 10 seconds, sent from a site in Victoria/Australia which has a standard time offset from LST with reference to UTC of -10 hours would be represented as: 20100701030010-10

4.3 Sensor Measurement Parameters

There can be one or more sensor measurements sent from within the one body of the message for this 'HS' message type. The sensor measurement parameters will specify the observed measurement for the identified sensor type, this measurement is also referenced by the observation timestamp section if it has been included within the message.

The sensor parameter reading is identified by a Sensor Tag which is then followed by the recorded reading value (which will be dependant on the sensor type). The sensor tag identifies the sensor type, sensor instance and measurement scheme, all of which are explained in the following tables.

The sensor instance is used to identify the sensor at sites that have more than one of the same sensor type connected to that site. Numbering of sensors commence at one (1) and are mandatory.

Measurement sampling definitions:

- Cumulative values increment continuously like a counter then restart at zero when the maximum range value (called rollover) is reached. The reporting increment size (step size) is equipment dependant.
- Instantaneous measurements are taken at a point in time and can vary up or down depending on the sensor precision and equipment range, readings can span into a new range (increase cycle count) when the maximum is value reached within that range.
- Sampled events are a measure value over a defined period of time and/or action.

Sensor Measurement data representation:

- Raw: As read measurement in non engineering units, in the form of an internal register or counter represented as an integer number.
- Translated: Measurement converted into appropriate engineering unit with precision and number of decimal places dependant on the sensor type and sensor measuring equipment at implementation time.

Syntax for this parameter is,

SENSORVALUE:

**{SensorType}_{SensorInstanceNum}-{SensorMeasurementType} =
{‘measured value’}**

Table 4 - Sensor Type Parameters

Sensor Type Class	Sensor Type Class Identifier	Unit of Measure	Decimal Accuracy	Description
Air Temperature	AT	Raw: N/A Translated: C	Raw: N/A Translated:1	Air Temperature measured in degrees Celcius
Barometric Pressure	BP	Raw: N/A Translated: hPa	Raw: N/A Translated:1	Barometric pressure measured in hecta Pascals.
Battery	B	Raw: N/A Translated: volts	Raw: N/A Translated:Up to 2	Observation measurement for equipment operating voltage level, typically this would the data acquisition device identified by sensor instance number one. If the sending device can measure the operating voltage of other sensors they could also be reported as different sensor instances.
Conductivity	C	Raw: N/A Translated: uS/cm	Raw: N/A Translated:1	Measured conductivity of micro seconds/cm.
Discharge	Q	Raw: N/A Translated: m3/s	Raw: N/A Translated:1	Measured discharge of water course in cubic metres per second.
Dissolved Oxygen	DO	Raw: N/A Translated: ppm	Raw: N/A Translated:1	Dissolved oxygen measure in parts per million.
Gas Pressure	GP	Raw: N/A Translated: mPa	Raw: N/A Translated:2	Gas pressure measured in mega Pascals.
pH	PH	Raw: N/A Translated: pH	Raw: N/A Translated:1	Measure pH.
Rainfall	R	Raw: N/A Translated: mm	Raw: N/A Translated:1	Precipitation measurement for cumulative counts and period based measurements.
Relative Humidity	RH	Raw: N/A Translated: %	Raw: N/A Translated:1	Relative humidity measure as a percentage
Solar Radiation	SR	Raw: N/A Translated: w/m2	Raw: N/A Translated:1	Solar radiation measured in watts/square metre.
Turbidity	TU	Raw: N/A Translated: ntu	Raw: N/A Translated:1	Turbidity measured in Nephelometric Turbidity Units
Water Level	H	Raw: N/A Translated: metres	Raw: N/A Translated: Up to 3	Observation measurement for attached water level sensor.
Water Temperature	WT	Raw: N/A Translated: C	Raw: N/A Translated:1	Water temperature measured in degrees Celcius.

Wind Direction	WD	Raw: N/A Translated: metres	Raw: N/A Translated:1	Wind direction measured in degrees.
Wind Gust	WG	Raw: N/A Translated: km/hr	Raw: N/A Translated:1	Wind gust measured in kilometres per hour.
Wind Run	WR	Raw: N/A Translated: km/hr	Raw: N/A Translated:1	Wind run measured in kilometres per hour.

Table 5 - Sensor Measurement Types

Sensor Type Identifier	Sensor Measurement Method	Sensor Report Format	Sensor Report Value Range	Description
0	Cumulative	Raw	0 to 2047	Integer accumulator with maximum value being 2047. Emulate USA ALERT version 1 protocol accumulator
1	Cumulative	Raw	0 to 9999	Integer accumulator with maximum value being 9999
2	Cumulative	Raw	0 to 99999	Integer accumulator with maximum value being 99999.
3	Cumulative	Raw	0 to 999999	Integer accumulator with maximum value being 999999.
4	Cumulative	Raw	0 to 9999999	Integer accumulator with maximum value being 9999999
6	Cumulative	Translated	99999.999	The maximum precision would be eight (8) with up to three (3) decimal places dependant on sensor type. e.g. Rainfall would typically be single decimal place, 9999.9.
10	Instantaneous	Raw	0 to 2047	Integer accumulator with maximum value being 2047 Emulate USA ALERT version 1 protocol accumulator
11	Instantaneous	Raw	0 to 9999	Integer accumulator

BoM Hydro Field Event Message Format

				with maximum value being 9999
12	Instantaneous	Raw	0 to 99999	Integer accumulator with maximum value being 99999.
13	Instantaneous	Raw	0 to 999999	Integer accumulator with maximum value being 999999.
14	Instantaneous	Raw	0 to 9999999	Integer accumulator with maximum value being 9999999.
16	Instantaneous	Translated	99999.999	The maximum precision would be eight (8) with up to three (3) decimal places dependant on sensor type. e.g. River height would typically be up to a millimetre measurement with 3 decimal places, 99999.999.
20	Sampled	Raw	0 to 2047	Integer accumulator with maximum value being 2047. Emulate USA ALERT version 1 protocol accumulator
21	Sampled	Raw	0 to 9999	Integer accumulator with maximum value being 9999..
22	Sampled	Raw	0 to 99999	Integer accumulator with maximum value being 99999.
23	Sampled	Raw	0 to 999999	Integer accumulator with maximum value being 999999.
24	Sampled	Raw	0 to 9999999	Integer accumulator with maximum value being 9999999
26	Sampled	Translated	99999999	The maximum precision would be eight (8) with up to three (3) decimal places.

4.4 Coded “HS=1” Message Examples

- a) Site consists of a single rainfall tipping bucket rain gauge reporting as an ALERT type accumulator (maximum 2048) with a value of 1055. No observation timestamp is required. Message length would be up to 30 characters.

:HS=1|I1=123456|R_1-0=1055|NN:

- b) Site consists of a single rainfall tipping bucket rain gauge reporting as an ALERT type accumulator with a value of 1055. No observation timestamp is required. A check message consisting of the current rainfall accumulator and equipment battery voltage would have a message length of up to 41 characters.

:HS=1|I1=123456|R_1-0=1055|B_1-10=139|NN:

- c) Site consists of a single rainfall tipping bucket rain gauge reporting as an ALERT type accumulator with a value of 1055. An observation timestamp is required to be sent in UTC format and time zone information isn't required. A check message consisting of the current rainfall accumulator and equipment battery voltage would have a message length of up to 59 characters.

:HS=1|I1=123456|T1=20100727030000|R_1-0=1055|B_1-10=139|NN:

- d) Site consists of a single rainfall tipping bucket rain gauge reporting as an ALERT type accumulator with a value of 1055. An observation timestamp is required to be sent in UTC format and time zone information isn't required. Message length is up to 48 characters.

:HS=1|I1=123456|T1=20100727030000|R_1-0=1055|NN:

- e) Site consists of two rainfall tipping bucket rain gauges both reporting as ALERT type accumulators with rainfall sensor number one having a value of 101 and rainfall sensor number two having a value of 201. No observation timestamp is required.

:HS=1|I1=123456|R_1-0=101|R_2-0=201|NN:

- f) Site consists of two rainfall tipping bucket rain gauges both reporting as ALERT type accumulators with rainfall sensor number one having a value of 101 and rainfall sensor number two having a value of 201. An observation timestamp is required to be sent in UTC format and time zone information isn't required.

:HS=1|I1=123456|T1=20100727030000|R_1-0=101|R_2-0=201|NN:

- g) Site consists of one rainfall tipping bucket rain gauge and one water level gauge both reporting as ALERT type accumulators with the rainfall sensor having a value of 1098 and the water level gauge having a value of 1051. An observation timestamp is

BoM Hydro Field Event Message Format

required to be sent in UTC format and time zone information is required, the time zone offset from UTC is plus ten (+10) hours.

:HS=1|I1=123456|T2=20100727030000+10|R_1-0=1098|H_1-10=1051|NN:

- h) Site consists of one rainfall tipping bucket rain gauge and one water level gauge both reporting as ALERT type accumulators with the rainfall sensor having a value of 1098 and the water level gauge having a value of 1051. An observation timestamp is required to be sent in local standard time and time zone information is required, the time zone difference between UTC and LST with reference to LST is minus ten (-10) hours.

:HS=1|I1=123456|T3=20100727130000-10|R_1-0=1098|H_1-10=1051|NN:

- i) Site initiated a maintenance message.

:HS=1|M=1|I1=123456|T1=20100727030000|R_1-0=101|B_1-10=139|NN:

- j) Site consists of one rainfall tipping bucket rain gauge and one water level gauge both reporting as translated values with the rainfall sensor having a value of '10.2' and the water level gauge height of '120.901'. An observation timestamp is required to be sent in local standard time and time zone information is required, the time zone difference between UTC and LST with reference to LST is minus ten (-10) hours.

:HS=1|I1=123456|T3=20100727130000-10|R_1-6=10.2|H_1-16=120.901|NN:

References

Draft (pilot) BoM Hydrology Message structure for field eventing equipment, Robert Thompson, September 2006.

End of document



Through the *Water Act 2007*, the Australian Government has given the Bureau of Meteorology responsibility for compiling and delivering comprehensive water information across Australia.

For more information

Visit our website at www.bom.gov.au/water

Send an email request to waterinfo@bom.gov.au



Australian Government

Bureau of Meteorology